Information Security

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Motivation


فَبِاَيِّ اٰلَآءِ رَبِّكُمَا تُكَذِّبٰنِ
تو (اے جن و انسان!) تم دونوں اپنے رب کی کون کون سی نعمتوں کو جھٹلاؤ گے؟
SURAH REHMAN 55:13

Recommended Books:-

- Stallings, W. (2022). Cryptography and network security: principles and practice. Pearson. ISBN: 9789332585225.

- Maymi, F., & Harris, S. (2016). CISSP All-in-One Exam Guide, 7th Edition. McGraw-Hill Education. ISBN: 9780071849272.

Security

Security is about protecting assets.

Security is the prevention of certain types of intentional actions from occurring in a system.

Security is the process of protecting assets from a wide range of threats, including intentional attacks, accidental events, and natural disasters.

- – These potential actions are ***threats***.
- – Threats that are carried out are ***attacks***.
- – Intentional attacks are carried out by an ***attacker***.
- – Objects of attacks are ***assets***.

CIA Triad



Confidentiality:

Ensuring that information is only accessible to authorized individuals.

Integrity:

Ensuring that information is accurate and complete and that it has not been tampered with.

Availability:

Ensuring that information is accessible to authorized users when they need it.

Goals of Security

- **Prevention**
  - Prevent attackers from violating security policy
- **Detection**
  - Detect attackers' violation of security policy
- **Recovery**
- – Stop attack, assess and repair damage
- **Survivability**
- – Continue to function correctly even if attack succeeds

Computer Security

- Computer security is the protection of computer systems and information from harm, theft, and unauthorized use. It is the process of preventing and detecting unauthorized use of computer systems.
- There are various types of computer security which is widely used to protect the valuable information of an organization. (Structural Tree)

  - ❑ Information Security
  - ❑ Application Security
  - ❑ Cyber Security
  - ❑ Network Security
  - ❑ etr.

Computer Security

Information security is securing information from unauthorized access, modification, and deletion.

Application Security is securing an application by building security features to prevent Cyber Threats such as SQL injection, DoS attacks, data breaches, etc.

Network security is achieved by securing both software and hardware technologies.

Cybersecurity is defined as protecting computer systems, which communicate over the computer networks

Information Security

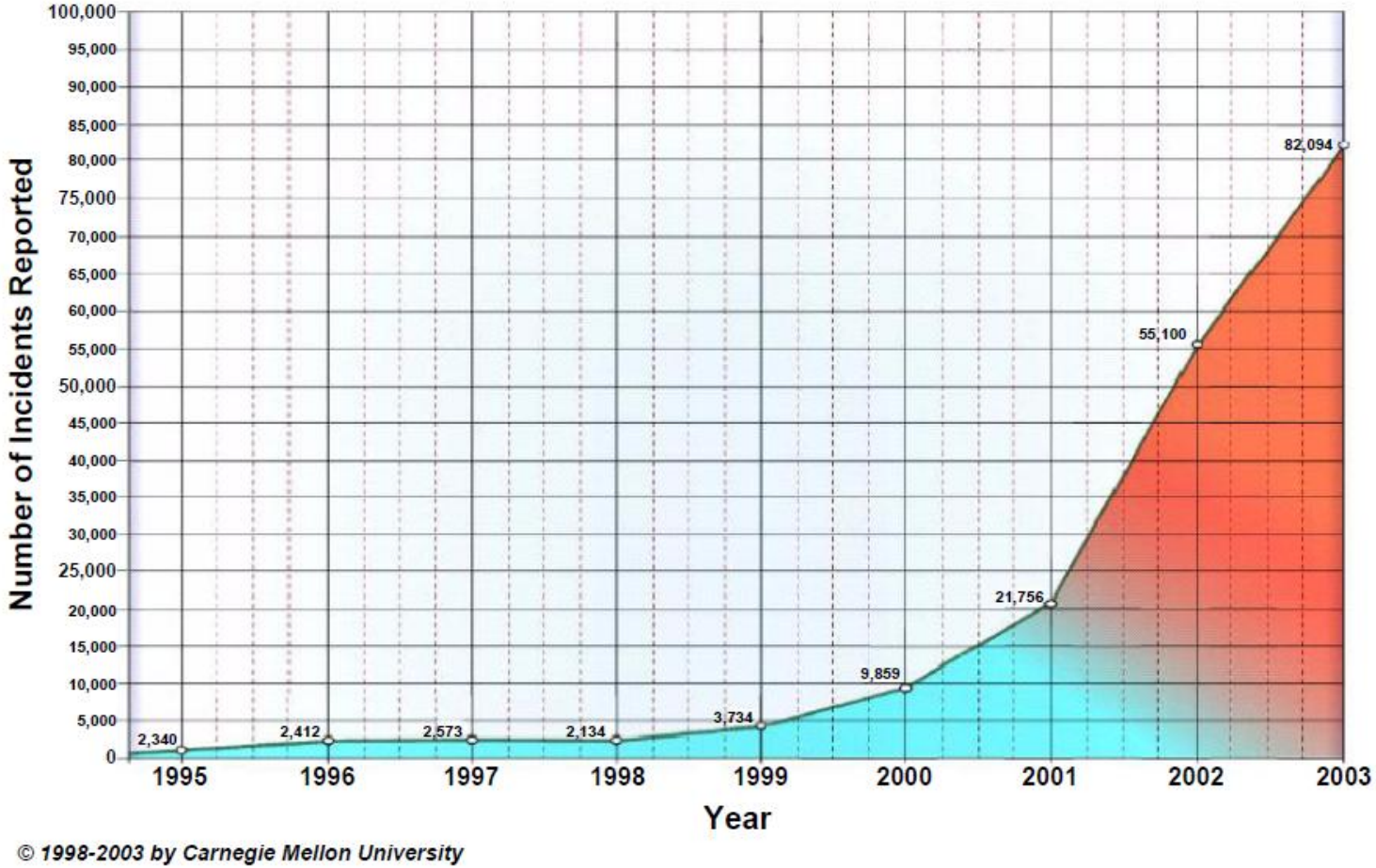
- Information Security is defending information from **unauthorized access, disclosure, disruption, modification, inspection, recording, or destruction.**

- It is a general term that can be used regardless of the data's form.

Security Incidents Reported to the CERT

Figure

Security incidents by year – CERT.ORG



Source: Fundamental Principles of Network Security, White Paper 101, Christopher Leidigh

Security Comes with Costs

- In reality, achieving perfect security not possible due to the costs associated with it.
- This cost can be measured not just in money, but also in complexity, time and efficiency.
- To make things secure, it is necessary to spend money, perform more procedures, and wait for these procedures to complete.
- Recommendation is to choose a scheme that has a certain amount of “cost” and an understood amount of security coverage

Jazak Allah Khair!